Continuous Time F.T.:

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad \& \quad x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

Properties:

9) Parseval's Relat^n:

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

Pf: $$\int_{-\infty}^{\infty} x(t) x^*(t) dt = \int_{-\infty}^{\infty} x(t) \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(\omega) e^{-j\omega t} d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(\omega) \underbrace{\int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt}_{X(\omega)} d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(\omega) X(\omega) d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

10) Duality Property:

$$x(t) \longleftrightarrow X(\omega)$$

$$\Rightarrow X(t) \longleftrightarrow 2\pi x(-\omega)$$

Pf: $$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} x(\tau) e^{-j\omega\tau} d\tau$$

$$\therefore X(t) = \int_{-\infty}^{\infty} x(\tau) e^{-jt\tau} d\tau = \int_{-\infty}^{\infty} x(\omega) e^{-jt\omega} d\omega \quad , \text{Let } -\omega = \omega'$$

$$= \int_{-\infty}^{\infty} x(-\omega') e^{j\omega' t} d\omega'$$

$$X(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} 2\pi x(-\omega) e^{j\omega t} d\omega$$

$$\therefore X(t) \longleftrightarrow 2\pi x(-\omega)$$

Ex. 1) $\delta(t) \longleftrightarrow 1$

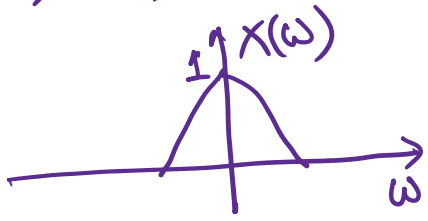
$$\therefore 2\pi\delta(-\omega) = 2\pi\delta(\omega)$$

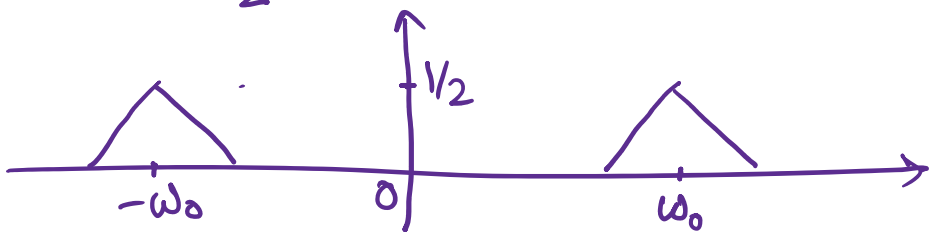
Ex. 1) $\delta(t) \longleftrightarrow 1$

$1 \longleftrightarrow 2\pi\delta(-\omega) = 2\pi\delta(\omega)$

2) $1 \cdot e^{j\omega_0 t} \longleftrightarrow 2\pi\delta(\omega-\omega_0)$

3) $x(t)\cos(\omega_0 t) \longleftrightarrow x(t)\dfrac{e^{j\omega_0 t}+e^{-j\omega_0 t}}{2} \longleftrightarrow \frac{1}{2}[X(\omega+\omega_0)+X(\omega-\omega_0)]$





11) $\underbrace{x(t)*h(t)}_{y(t)} \longleftrightarrow X(\omega)H(\omega) = Y(\omega)$

Pf: $\displaystyle\int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau = \int_{-\infty}^{\infty} x(\tau)\frac{1}{2\pi}\int_{-\infty}^{\infty} H(\omega)e^{-j\tau\omega}\cdot e^{j\omega t}\,d\omega\,.d\tau$

$\displaystyle = \frac{1}{2\pi}\int_{-\infty}^{\infty} H(\omega)e^{j\omega t}\,d\omega \underbrace{\int_{-\infty}^{\infty} x(\tau)e^{-j\omega\tau}d\tau}_{X(\omega)}$

$\displaystyle y(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \underbrace{H(\omega)X(\omega)}_{Y(\omega)}e^{j\omega t}\,d\omega$

$\therefore\ x(t)*h(t) \longleftrightarrow X(\omega)H(\omega)$

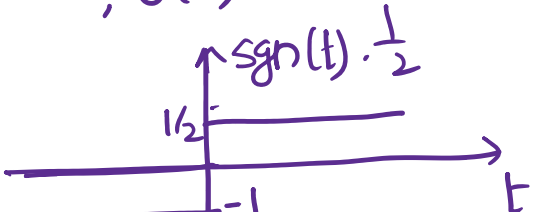
12) $x(t)h(t) \longleftrightarrow \frac{1}{2\pi}X(\omega)*H(\omega)$

Pf: Hint: $\displaystyle\frac{1}{2\pi}\int_{-\infty}^{\infty} X(\rho)H(\omega-\rho)\,d\rho =$

13) $\underbrace{\displaystyle\int_{-\infty}^{t} x(\tau)d\tau}_{} \longleftrightarrow ?$

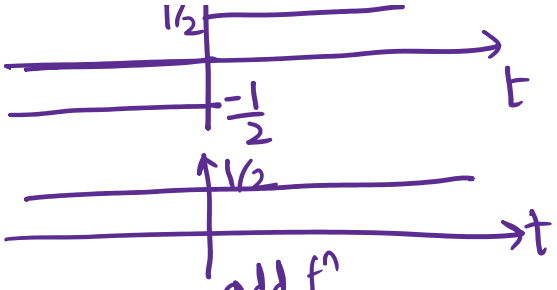
$x(t)*u(t) \longleftrightarrow X(\omega)U(\omega)$, $U(\omega)$?

$\displaystyle U(\omega) = \int_{0}^{\infty} e^{-j\omega t}dt =$



$u(t) = \frac{1}{2}[\mathrm{sgn}(t)+1]$

$$u(t) = \frac{1}{2}[\underbrace{\mathrm{sgn}(t)+1}_{2\pi\delta(\omega)}]$$



odd f^n

$$\mathrm{sgn}(t) \longleftrightarrow ?$$

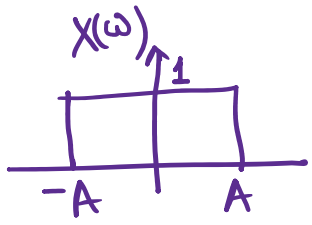
$$f(t) = \frac{1}{t} \longleftrightarrow F(\omega) = \int_{-\infty}^{\infty} \frac{1}{t} e^{-j\omega t}\, dt = \int_{-\infty}^{\infty} \frac{\cos(\omega t)}{t} - j\,\frac{\sin(\omega t)}{t}\, dt$$

$$= -j\int_{-\infty}^{\infty} \frac{\sin(\omega t)}{t}\, dt$$

$$F(\omega) = -j\pi \int_{-\infty}^{\infty} \frac{\sin(\omega t)}{\pi t}\, dt \Big\} = -j\pi,\ \omega > 0 \quad = j\pi,\ \omega < 0$$

$x(t)$

$$\frac{\sin At}{\pi t} \longleftrightarrow$$



$$X(\omega) = \int_{-\infty}^{\infty} \frac{\sin At}{\pi t} e^{-j\omega t}\, dt$$

$$X(0) = \int_{-\infty}^{\infty} \frac{\sin At}{\pi t}\, dt = 1$$

$$\therefore F(\omega) = -j\pi\,\mathrm{sgn}(\omega) \Rightarrow \frac{1}{t} \longleftrightarrow -j\pi\,\mathrm{sgn}(\omega)$$

$$\Rightarrow \cancel{+}j\cancel{\pi}\,\mathrm{sgn}(t) \longleftrightarrow \frac{\cancel{+}2\cancel{\pi}}{\omega}$$

$$\therefore \mathrm{sgn}(t) \longleftrightarrow \frac{2}{j\omega} \qquad , u(t) = \frac{1}{2}\mathrm{sgn}(t) + \frac{1}{2}$$

$$\Rightarrow U(\omega) = \frac{1}{j\omega} + \pi\delta(\omega)$$

$$\therefore \int_{-\infty}^{t} x(\tau)\, d\tau \longleftrightarrow \frac{X(\omega)}{j\omega} + \pi X(0)\delta(\omega)$$